

Lightweight IoT-edge framework with attention-enhanced GRU for closed-loop real-time indoor environmental monitoring and prediction

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ARTICLE INFO

Keywords:

Internet of things
Indoor environmental monitoring
Edge computing
Prediction
Real-time
GRU-SAM

ABSTRACT

Accurate real-time indoor environmental monitoring and prediction is essential for proactive control to protect occupant health and comfort. While prior studies have demonstrated real-time monitoring and AI-based prediction, real-time operation is often treated as near-real-time and evaluated using stored datasets rather than as a continuously operating, closed-loop streaming inference process. To address this practical and methodological gap, this study presents the initial phase of the Information and Communication Technology-enabled indoor air (ICTAir) framework, a real-time approach that integrates lightweight Internet of Things (IoT) technologies with hybrid deep learning (DL) model for continuous monitoring and prediction under a live closed loop deployment, while maintaining superior prediction accuracy with moderate inference-time and memory requirements compared to statistical and traditional machine learning approaches. ICTAir integrates low-cost sensors, a microcontroller, and a cloud back end, and employs an attention-enhanced gated recurrent unit with a simplified attention mechanism (GRU-SAM). The proposed GRU-SAM model is benchmarked against statistical baselines, traditional machine-learning models, and deep-learning alternatives, demonstrating consistently superior predictive accuracy. With a 1-h training window and a 5-min prediction horizon, GRU-SAM achieves low error (e.g., temperature RMSE = 0.22). Calibration experiments confirm sensor reliability and indicate a marked improvement in PM_{2.5} measurements after calibration. During deployment, temperature frequently exceeded 30 °C and relative humidity surpassed 60%, while PM_{2.5} and PM₁₀ often crossed 35 and 50 µg/m³, respectively. Continuous operation achieved end-to-end system latency within 1 min while maintaining 97.7% streaming inference reliability. The proposed framework provides a practical, deployment-oriented foundation for future autonomous indoor environmental control systems.

1. Introduction

Indoor environmental monitoring, often referred to as indoor environmental quality (IEQ) assessment, concerns the comprehensive characterization of physical and chemical conditions within enclosed spaces that influence occupants' comfort, health, and overall well-being. These conditions primarily include thermal factors such as temperature and relative humidity, which govern thermal comfort and perceived indoor acceptability, as well as airborne pollutants such as particulate matter, which are central indicators of indoor air quality and health risk. Its importance is emphasized by findings that urban populations spend more than 90% of their time indoors [1]. The COVID-19 pandemic has further heightened awareness of indoor environmental health, as extended periods of indoor occupancy have increased concerns regarding air quality. Prolonged exposure makes indoor environment a

critical factor in determining human health, productivity, and overall quality of life [2]. Elevated concentrations of indoor air parameters have been linked with a range of adverse outcomes, including cardiovascular and respiratory diseases, as well as impacts on mental and perceptual well-being [3]. In some cases, indoor pollutant concentrations can be up to 100 times higher than those found outdoors [4]. Indoor air can contain pollutant levels 2 to 5 times higher than outdoor air (U.S. Environmental Protection Agency 2025), while approximately 6.7 million premature deaths annually are attributed to ambient and household air pollution [5].

In the past, indoor air pollutant concentrations were measured using distinct sampling and laboratory-based techniques [6,7]. While these methods provided accurate data, they were often time-consuming, required specialized expertise, and lacked real-time responsiveness. With recent advancements in sensor technology, a growing body of

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<https://doi.org/10.1016/j.measen.2026.101994>

Received 20 October 2025; Received in revised form 26 January 2026; Accepted 10 February 2026

Available online 10 February 2026

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research has shifted toward the development of low-cost, energy-efficient sensor-based systems for indoor environmental monitoring. These systems have become increasingly viable for deployment in a wide range of indoor environments, enabling continuous data collection and user-oriented feedback. The convergence of Internet of Things (IoT) technologies, advanced sensors, and artificial intelligence (AI) techniques has further propelled this transition, allowing real-time monitoring, classification, and interpretation of key indoor environmental indicators.

Real-time indoor environmental monitoring is essential for the early detection of indoor environmental hazards such as particulate matter ($PM_{2.5}$ and PM_{10}), temperature fluctuations, and relative humidity imbalances [8,9]. Timely identification of such conditions enables prompt corrective action by occupants or automated building systems. Beyond real-time monitoring, the ability to predict trends under dynamic indoor conditions has become increasingly vital. Real-time predictions allow for the anticipation of pollution events and support proactive decision-making. Furthermore, the integration of deep learning (DL) algorithms into indoor environmental monitoring frameworks significantly enhances predictive accuracy. This synergy between real-time monitoring and predictive modeling lays the groundwork for proactive indoor environmental control systems.

Despite growing interest in DL-driven indoor environmental prediction, limited attention has been given to ensuring that such models remain computationally feasible for real-time deployment on resource-constrained IoT edge platforms. Deploying deep learning-based prediction models in Internet of Things (IoT) environments presents practical challenges due to the limited computational resources of edge devices [10–12]. Many state-of-the-art deep learning architectures used for time-series prediction are computationally intensive and memory-demanding, often requiring high-performance CPUs, GPUs, or cloud-based infrastructures for stable real-time inference. When such models are executed on resource-constrained edge platforms, they can lead to increased inference latency and unreliable real-time performance (Aminu et al., 2025; [13,14]). These limitations are particularly critical for closed-loop indoor environmental monitoring systems, where predictions must be generated within fixed sampling intervals to enable timely responses, thereby necessitating lightweight inference frameworks that operate under bounded memory and computational budgets using fixed-length sliding windows.

This study focuses on real-time monitoring and short-term prediction of key indoor environmental parameters, namely indoor temperature, relative humidity, $PM_{2.5}$, and PM_{10} , using a lightweight IoT sensing architecture integrated with deep learning models. In this work, we focus on predicting indoor environmental parameters to support two complementary objectives: (i) thermal comfort evaluation at the parameter level using temperature and relative humidity, and (ii) indoor air quality assessment based on particulate matter exposure ($PM_{2.5}$ and PM_{10}). The selection of thermal and air quality parameters is motivated by their direct and complementary influence on human comfort, health, and effective control. Temperature and relative humidity are the primary physical drivers of thermal comfort and are strongly associated with occupant satisfaction, productivity, and perceived indoor acceptability. In parallel, particulate matter ($PM_{2.5}$ and PM_{10}) represents one of the most critical indicators of indoor air quality, with well-established links to respiratory and cardiovascular health risks. From a system perspective, these variables are directly actionable through common building control mechanisms, such as split air-conditioning systems for regulating thermal conditions and ventilation or HEPA filtration units for particulate removal.

In this context, lightweight IoT refers to a resource-efficient system design that combines a minimal set of low-cost sensors with the deployment of a deep learning prediction model on a compact edge platform (Raspberry Pi), operating under limited computational resources rather than relying on dense sensing and computationally expensive analytics. By prioritizing architectural simplicity and low

computational overhead, the proposed framework enables continuous, closed-loop prediction on the edge while maintaining a 1-min end-to-end latency. In this work, thermal comfort is represented at the parameter level through temperature and relative humidity, which are widely recognized as primary determinants of perceived thermal conditions. Composite thermal comfort indices (e.g., PMV/PPD) and aggregated air quality indices that integrate multiple pollutants are beyond the scope of this initial phase. Instead, the proposed framework provides accurate, real-time parameter-level predictions that can serve as a foundational layer for future comfort-index estimation and closed-loop indoor environmental control applications.

2. Related work

The literature review is structured into two key thematic categories: (1) Indoor environmental monitoring, (2) Indoor environmental prediction.

2.1. Indoor environmental monitoring

Numerous studies have highlighted the effectiveness of IoT technologies for comprehensive indoor environmental monitoring. These integrated systems facilitate real-time data acquisition, pollutant classification, and intelligent decision-making processes [15]. IoT-enabled platforms support continuous environmental monitoring and provide actionable feedback. Various systems have been designed to track multiple environmental parameters in real-time while employing machine learning models to analyze the impact of human activity and indoor environmental variations [16]. Some implementations utilize wireless sensor nodes to detect a range of gaseous pollutants and transmit the collected data to centralized servers for real-time processing and visualization [17]. Wireless sensor networks (WSNs), particularly those employing low-power ZigBee communication modules, have also been used to detect volatile organic compounds (VOCs) and synchronize with cloud-based platforms for remote analysis [18]. Additional innovations include Arduino-based architectures equipped with edge computing capabilities for local data processing [19], as well as compact, modular sensor systems designed to dynamically display pollutant fluctuations with visual feedback mechanisms [20]. These IoT-driven indoor environmental monitoring solutions demonstrate strong potential for real-time observation, emphasize their role in enhancing indoor environmental quality and supporting energy-efficient building management practices [21–23].

Several studies have implemented sensor-based indoor environmental monitoring systems over varying data collection periods. Short-term experiments include 1-h monitoring using wireless sensor networks (WSNs) for pollutant tracking [24], integrating metal oxide semiconductor gas sensors with cloud platforms [25], and employing long-range sensor networks for multi-pollutant tracking [26]. Medium-term studies range from one-day real-time health monitoring to seven-day monitoring of ten indoor pollutants in multiple environments [27], as well as two-week experiments analyzing the effects of relative humidity and temperature using sensor-generated data. Long-term deployments have monitored indoor environment continuously for one year to enable immediate environmental assessment [28], collected occupant comfort feedback via wireless indoor environment loggers over three months [29,30], implemented portable multi-sensor systems for a month-long assessment [31,32], and carried out 60-h WSN-based indoor environmental monitoring in academic settings [33]. These studies demonstrate a wide range of monitoring durations, from highly localized short-term measurements to extended multi-month deployments, reflecting diverse application goals and system designs in indoor environmental research.

2.2. Indoor environmental prediction

Predicting indoor environmental parameters plays a critical role in anticipating pollutant concentrations and supporting the design of effective mitigation strategies. Accurate prediction enables timely public health interventions and informs the development of intelligent environmental control systems. In recent years, various machine learning (ML) algorithms, including support vector regression, multiple linear regression, random forest, and artificial neural networks (ANN) have been employed to predict air quality, yielding promising outcomes

[34–38]. Despite their effectiveness in capturing static relationships among variables, traditional ML models often struggle to account for temporal dependencies inherent in time-series indoor environmental data. To overcome this limitation, DL techniques have emerged as superior alternatives, offering advanced capabilities to learn complex temporal and spatial patterns. Among them, DL-based methods enhanced with attention mechanisms are particularly effective in improving predictive model performance. Numerous studies have broadened the application of DL techniques for predicting indoor environment, utilizing datasets collected over varying durations from either

Table 1
Summary of deep learning-based studies for predicting indoor environment.

Reference	Models	Parameters	Real-time		Description	Limitation
			M	P		
Sharma et al. [39]	LSTM	CO ₂ , PM _{2.5}	✓	×	Modified long short-term memory (LSTM) networks were trained on multi-week datasets to improve prediction accuracy, achieving up to 96% accuracy.	Real-time prediction is implemented at the server level rather than on the edge device.
Mtibaa et al. [40]	LSTM	Temperature	✓	×	Sequence-to-sequence LSTM architectures have shown strong performance in forecasting multi-zone indoor temperatures, achieving up to a 50% reduction in prediction error across forecast horizons of up to 6 h.	The LSTM-based framework is evaluated offline, with closed-loop streaming feasibility not explicitly examined and the analysis limited to a single modeling approach.
Ahn et al. [41]	GRU, LSTM	CO ₂ , Fine dust, Temperature, Humidity, light quantity, VOCs	✓	×	Both GRU and long short-term memory (LSTM) architectures have been employed to model long-term indoor environment dynamics over a seven-month period, achieving an accuracy of 84.69%.	The work emphasizes server-based air quality prediction with offline evaluation; comparisons are limited to selected models
Khazaei et al. [42]	MLP neural network	CO ₂	✓	×	Multilayer perceptron (MLP) has been applied to predict CO ₂ levels multiple time steps ahead by incorporating contextual variables such as temperature and humidity.	An MLP-based CO ₂ predictor is evaluated offline; comparisons are limited to feedforward models, and edge-hardware feasibility, and computational overhead are not examined.
Wang et al. [22,23]	GRU	Humidity, Temperature, Light	✓	×	A GRU-based model leveraging spatially correlated time-series data from sensor nodes demonstrated improved predictive accuracy for indoor air pollutants, achieving approximately 14% and 5% reductions in MAE and RMSE, respectively.	A GRU-based deep learning prediction is demonstrated with training conducted offline, while computational cost and system inference are not evaluated.
Zhang et al. [43]	LSTM	VOCs	×	×	LSTM-based models have been evaluated in both controlled laboratory conditions and real-world classroom environments for forecasting volatile organic compounds (VOCs).	The framework is validated using controlled chamber and classroom datasets with offline evaluation; edge-hardware constraints and computational complexity are not explicitly assessed.
Mohammadshirazi et al. [44]	Rolling Average, Random Forest, Gradient Boosting, and LSTM	CO ₂ , NO ₂ , O ₃ , PM ₁ , PM _{2.5} , PM ₁₀ , formaldehyde, total volatile organic compounds (TVOCs)	✓	×	The predictive performance of four algorithms was evaluated for forecasting eight indoor pollutant concentrations using low-cost sensors in a commercial building, with LSTM explaining 90% of the variance in time-series data for most pollutants.	The study focuses on offline, hourly-resolution prediction in a single building; end-to-end real-time latency is not explicitly evaluated.
Nurcahyanto et al. [45]	RNN, LSTM	PM ₁₀ , Humidity, Temperature	✓	×	A multilevel RNN framework was developed for PM ₁₀ prediction in a cleanroom environment using IoT data, achieving a significantly lower MSE (0.1296) compared to the LSTM model (0.2592), indicating superior predictive performance.	The multilevel RNN is validated in a controlled cleanroom setting; quantitative evaluation of computational complexity and end-to-end real-time closed-loop streaming remains outside the scope.
Ma et al. [46]	BiLSTM	Air velocity, temperature, humidity	✓	×	Proposed real-time thermal comfort prediction models evaluated at 10-, 30-, and 60-min time scales in campus buildings, achieved superior accuracy (RMSE = 0.0551, MAE = 0.0543, R ² = 0.9816 at 10-min scale)	The study targets PMV prediction using Bi-LSTM models evaluated in a workstation environment; edge-hardware constraints and continuous closed-loop streaming deployment are not explicitly assessed.
Kumar & Kurian [47]	SVR, Tree, Ensemble tree, ANN	Air velocity, temperature, humidity	✓	×	Developed framework for real-time prediction of PMV, SET, and 7-point thermal sensation using IoT, achieving accuracy (PMV: RMSE = 0.0016, R ² = 1.00; SET: RMSE = 0.0357, R ² = 0.9998; TS accuracy = 98.7%).	The work emphasizes real-time thermal comfort prediction; however, system-level continuous end-to-end streaming latency and computational complexity are not discussed.
Yildiz & Sucuoglu [48]	ML Models, LSTM, GRU	Humidity, temperature, CO, TVOCs, eCO ₂ , PM _{2.5}	✓	×	Developed IoT-based multi-gas air quality monitoring and short-term AQI forecasting system, where a GRU model achieved R ² > 0.93 with inference latency below 130 ms and over 91% accuracy.	The work focuses on real-time AQI forecasting and classification, while parameter-level prediction and continuous closed-loop streaming inference remain outside its scope.

M: monitoring; P: prediction; ✓: supported; ×: not supported

weather monitoring stations or custom-built sensing systems, as summarized in Table 1. In Table 1, real-time prediction (P) refers to a continuously operating inference process in which live sensor data are directly ingested by the prediction model after each acquisition cycle.

Indoor environment time-series data often exhibit long quasi-stationary periods under steady occupancy and ventilation conditions, but they can change abruptly during short-lived events such as cleaning activities, occupant movement, window opening, or air-conditioning operation [49,50]. These transient episodes may disproportionately influence near-future concentrations, yet their temporal patterns can be diluted when sequence models implicitly average information across the full input window. Attention mechanisms address this issue by learning to assign higher weights to the most informative time steps, allowing the predictor to focus on event-driven deviations while retaining the broader temporal context [51–53]. This characteristic is particularly suitable for indoor environmental forecasting, where a small subset of recent observations often governs the next-step dynamics.

The attention mechanism, inspired by the human cognitive process of focusing on specific stimuli, was initially introduced in computer vision research to reduce computational load by selectively processing important regions of an image rather than the entire input frame [54]. Its adoption rapidly expanded, particularly in natural language processing (NLP), incorporated attention into neural machine translation models [55]. Since then, attention mechanisms have become integral to DL models across a wide range of applications including object detection [56], medical image segmentation, time-series forecasting [57], NLP tasks [58], and question-answering systems [59], among others [60–62]. In the domain of environmental forecasting, the attention mechanism has recently gained traction for air quality prediction tasks. For example, a study [63] introduced a variational autoencoder enhanced with directed attention for forecasting pollutants such as NO₂, O₃, SO₂, and CO, demonstrating superior accuracy across six evaluation metrics. An LSTM-based attention model [64] was developed and tested on Beijing air quality dataset, outperforming traditional baselines like ARIMA, DeepST, and GC-LSTM in terms of RMSE and R². Similarly, in the work [65] presented a sequence-to-sequence attention-based framework that replaced the RNN encoder with a pure attention mechanism, yielding improved PM_{2.5} forecasting performance at two urban monitoring stations. In another effort [66], a dual-LSTM architecture was proposed, integrating an attention-enhanced LSTM for multi-factor forecasting and combining it with a seq2seq model and XGBoost regression to produce final predictions, showcasing improved modeling of pollutant trends.

2.3. Motivation, research gaps, challenges and research questions

While indoor environment encompasses a wide range of pollutants including carbon dioxide (CO₂), volatile organic compounds (VOCs), and formaldehyde, the proactive control of thermal comfort parameters (temperature and relative humidity) and particulate matter (PM_{2.5} and PM₁₀) is directly linked to occupant comfort, productivity, and respiratory health. Prior studies have reported AI-based prediction for indoor environmental monitoring and often describe their systems as “real-time.” However, real-time operation is interpreted as near-real-time or periodic inference, where models are evaluated on stored data streams, batch-processed windows, or executed on external servers or workstations. Consequently, prediction is frequently treated as an offline or analytical capability rather than as a continuously operating, end-to-end streaming inference process that is temporally synchronized with on-device sensing and control cycles. Unlike prior indoor environmental studies that primarily emphasize offline prediction accuracy, this work contributes an end-to-end, latency-aware IoT-deep learning framework that jointly addresses sensor calibration, real-time streaming continuity, lightweight attention-based forecasting, and edge-deployment feasibility.

To address these practical and methodological gaps, the present

work implements real-time prediction as a fully integrated, closed-loop process in which live sensor data are acquired, calibrated, stored, and directly ingested by a deep learning model deployed on a Raspberry Pi-based edge platform. By combining cost-effective sensors, Internet of Things (IoT) infrastructure, and a lightweight deep learning model, this study evaluates the capability to monitor temperature, relative humidity, PM_{2.5}, and PM₁₀ in real-time and to predict their short-term variations with high accuracy. After each acquisition cycle, short-term predictions are generated and made immediately available, ensuring continuous, low-latency access to forecasted parameter-level values suitable for control integration. Existing indoor environmental control practices nevertheless remain largely reactive, relying on manual interventions such as opening windows, adjusting air conditioners, or activating air purifiers only after threshold exceedances have occurred, which may expose occupants to unhealthy indoor conditions before corrective actions take effect. The resulting continuously updated predictions are explicitly designed to support future integration with automatic indoor environmental control systems.

Despite growing interest, several challenges hinder the practical implementation of real-time predictive indoor environmental control systems. Technical issues such as delayed system responses, sensor inaccuracies, and reliance on manual operation reduce the effectiveness of current solutions. Human factors also contribute, for instance, neglecting to turn off air-conditioning units can increase energy consumption and cause inefficient indoor environmental control. Furthermore, real-time deployment of predictive models often faces computational and data-handling constraints. These challenges emphasize the need for robust, data-driven predictive frameworks that can operate efficiently in real-time and form the basis for future autonomous indoor environmental control systems.

To address these challenges, this study seeks to answer the following research questions:

- Can a lightweight IoT sensing architecture be designed to support real-time monitoring and prediction as a fully integrated, closed-loop process for parameter-level indoor environmental variables (temperature, relative humidity, PM_{2.5}, and PM₁₀), while consistently satisfying an end-to-end latency constraint of 1 min?
- How effectively does the proposed hybrid deep learning model (GRU-SAM) predict short-term variations in indoor environmental parameters compared with existing baseline models?
- What is the optimal configuration of training and prediction window lengths for the GRU-SAM model to maximize short-term prediction accuracy across thermal comfort parameters (temperature and relative humidity) and particulate matter (PM_{2.5} and PM₁₀)?
- To what extent does sensor-level calibration enhance the measurement accuracy and operational reliability of low-cost indoor air quality monitoring systems in real-world deployment scenarios?

The remainder of this article is organized as follows. Section 3 describes the research methodology, including the experimental environment, hardware configuration, sensor calibration procedures, cloud integration, and the real-time monitoring and prediction protocols. Section 4 presents the results and discussion, structured into four main parts: (1) statistical volatility and temporal dynamics of indoor environmental variables; (2) comparative accuracy evaluation of the proposed GRU with simplified attention mechanism (GRU-SAM) model against classical statistical, machine-learning, and deep-learning baselines for short-term prediction of thermal comfort parameters and particulate matter concentrations; (3) optimization analysis of training and prediction window sizes for the GRU-SAM model; and (4) evaluation of real-time system latency, streaming inference reliability, and inference-time memory footprint under edge deployment. Finally, Section 5 concludes the work by summarizing the key findings, contributions and limitations.

3. Methodology

3.1. Data collection

For the data collection, a research laboratory located in the Engineering Department at Dong-A University (35°06'59" N, 128°58'01" E), Busan, South Korea, was selected as the primary experimental site. Situated on the fourth floor of a rectangular academic building, the laboratory has one facade facing a road and the other adjacent to a small mountain. The space covers a floor area of 60 m² with a total volume of 154 m³ and typically accommodates 14 researchers during standard working hours. It is equipped with two sliding windows that enable natural ventilation and two ceiling-mounted split air-conditioning units with independent control panels located near the entrance. Additional features include 12 ceiling-mounted luminaires and a single main door for entry and exit, resulting in a dynamic indoor environment for monitoring experiments.

This laboratory was deliberately chosen as the initial testbed for the ICTAir project because it represents a highly occupied, day-to-day academic office environment with sustained human activity, frequent short-term indoor events, and regular operation of split air-conditioning and mixed-mode ventilation (window opening). Such characteristics provide a practical and information-rich setting to identify the need for proactive indoor environmental control under realistic usage conditions and to validate the feasibility of continuous, real-time sensing and closed-loop streaming inference on low-cost edge hardware. However, indoor environmental dynamics may vary across buildings and climatic contexts; accordingly, subsequent phases of this project will extend validation to additional office and laboratory spaces and across different seasons to assess generalizability. The detailed characteristics of the laboratory are summarized in Fig. 1.

3.1.1. Hardware

In the proposed indoor environmental monitoring framework, we selected low-cost sensors and a controller to measure key indoor environmental indicators. We utilized the PM2008M sensor, which employs a laser scattering technique to accurately detect PM_{2.5} and PM₁₀ concentrations up to 1000 µg/m³. To measure thermal comfort parameters (temperature, relative humidity), the DHT-22 sensor was selected for its dual capability in capturing temperature (−40 °C to +80 °C) and relative humidity (0%–100%) using capacitive and thermistor-based sensing technologies, respectively. These sensors were chosen based on their measurement precision, ease of calibration, compatibility with Raspberry Pi for seamless hardware integration, availability, and cost-effectiveness.

The monitoring system focuses on four essential indoor

environmental parameters: two comfort variables, temperature and relative humidity, and two particulate matter indices, PM_{2.5} and PM₁₀. These indicators were selected due to their prevalence in indoor environments and their direct impact on occupant health and comfort. Moreover, each parameter is manageable through automated systems, such as air conditioners for regulating temperature and relative humidity, and HEPA filtration units for effective particulate matter control. Raspberry Pi 4B was selected as the central control unit for the sensor array due to its lightweight features, robust Python programming environment, built-in Wi-Fi capabilities, and compatibility with future sensor expansion. The device operates continuously, 24 h a day, using a stable power supply to ensure uninterrupted real-time data acquisition. This design is intentionally flexible, allowing for the easy substitution in an open configuration to facilitate the second phase of our research. For later deployment, the system will be housed in a custom-designed 3D-printed enclosure for practical indoor use.

Upon activation, the device automatically connects to the local Wi-Fi network, initiates data collection from the attached sensors, and transmits the gathered data to the ThingSpeak cloud platform for secure storage and real-time visualization on designated channel fields. Fig. 2 illustrates the initial ICTAir system architecture, comprises a Raspberry Pi 4B (1) with an extension kit (2) and breadboard, functioning as the central processing unit, connected to the DHT-22 sensor (3) and the PM2008M sensor (4). The display terminal (5) shows live sensor data successfully transmitted to the cloud, including PM_{2.5}, PM₁₀, temperature, and relative humidity. This configuration enabling continuous data acquisition and seamless cloud communication.

3.1.2. Sensor data calibration

To ensure the accuracy and reliability of the monitoring system, a sensor calibration process was conducted using the Temtop M2000 as the reference instrument. The Temtop M2000, certified under ISO quality inspection standards, provides precise measurements, and supports data logging and export for analytical validation. During calibration, all sensors were co-located with the reference device, and simultaneous readings were recorded for 500 data points. This number of calibration samples was selected to ensure sufficient coverage of short-term environmental variability while maintaining stable operating conditions, in line with common practices for laboratory-based indoor environmental sensor calibration. The raw outputs from the deployed sensors were compared with the Temtop M2000 readings to identify systematic deviations. Linear regression analysis was then applied to correct the low-cost sensor outputs. This approach followed standard laboratory calibration procedures and adhered to indoor air quality monitoring guideline [67–69], ensuring consistent transformation of raw data into accurate environmental measurements.

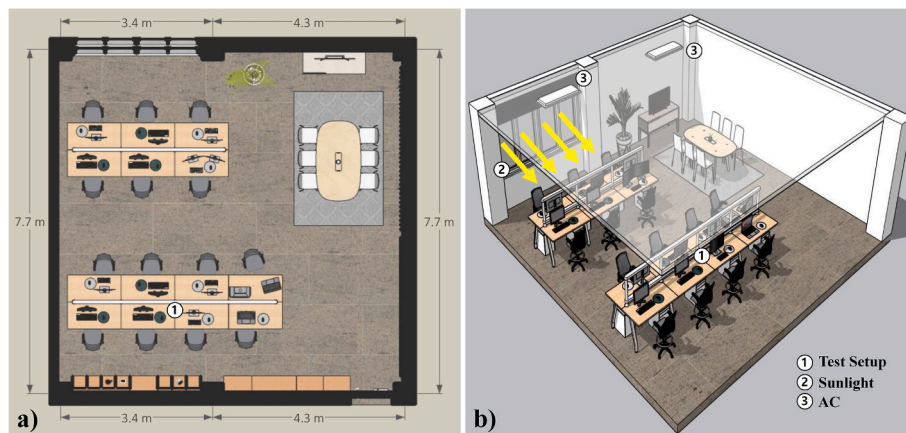


Fig. 1. Experimental site: (a) 2D layout of the laboratory with furniture and fixtures; (b) 3D representation of the laboratory, showing the location of the experimental setup, air-conditioning (AC), sunlight, and seating arrangement.

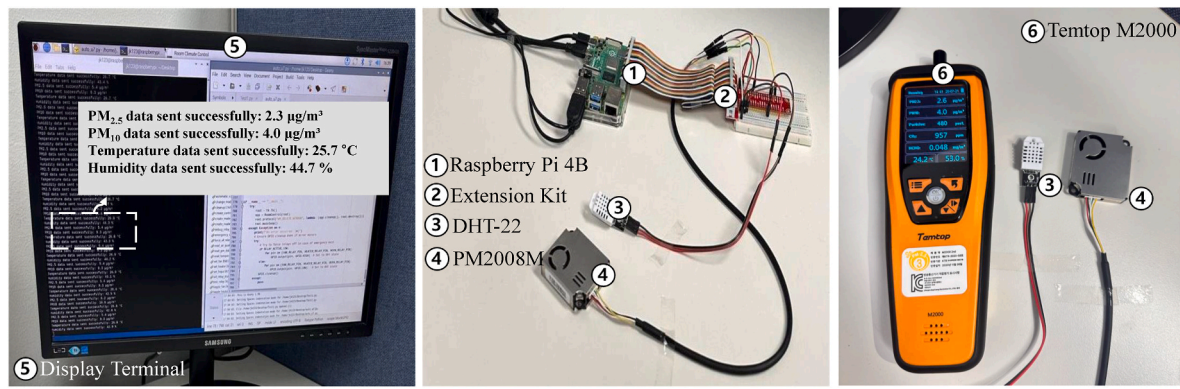


Fig. 2. Overview of the initial ICTAir framework components: (1) Raspberry Pi 4B; (2) extension kit; (3) DHT-22 sensor; (4) PM2008M sensor; (5) display terminal; and (6) Temtop M2000 for calibration.

The sensors and the reference device were stabilized for 1 h to reduce fluctuations and ensure consistent air quality readings [70]. By positioning the sensors in close proximity to the Temtop M2000, we ensured exposure to identical environmental conditions, allowing for direct comparison. Fig. 2 illustrates this calibration setup with Temtop M2000 (6). Corrections were integrated into the Python (version 3.11.2) code running on the Raspberry Pi, ensuring that all real-time outputs aligned with the reference standards provided by the Temtop M2000. To visually validate the calibration performance, scatter correlation plots between sensor measurements and reference values were generated before and after calibration for temperature, relative humidity, PM_{2.5}, and PM₁₀, as shown in Fig. 3. Each subplot presents raw and calibrated sensor readings against the corresponding reference measurements,

together with the fitted linear regression lines and the 1:1 reference line. In the regression equations, S denotes the sensor measurement, while R represents the reference measurement.

Following the calibration process, post-calibration tests were carried out to evaluate the accuracy and stability of the sensor readings. As shown in Table 2, further analysis of the calibration and post-calibration tests of our indoor environmental sensors revealed high precision and stability across parameters. Prior to calibration, the PM_{2.5} readings showed noticeable deviations and fluctuations relative to the reference values, with a root mean square error (RMSE) of 8.70, mean absolute error (MAE) of 8.29, and mean squared error (MSE) of 7.62. To address these discrepancies, a calibration procedure was applied to align the sensor outputs more closely with those of the reference device.

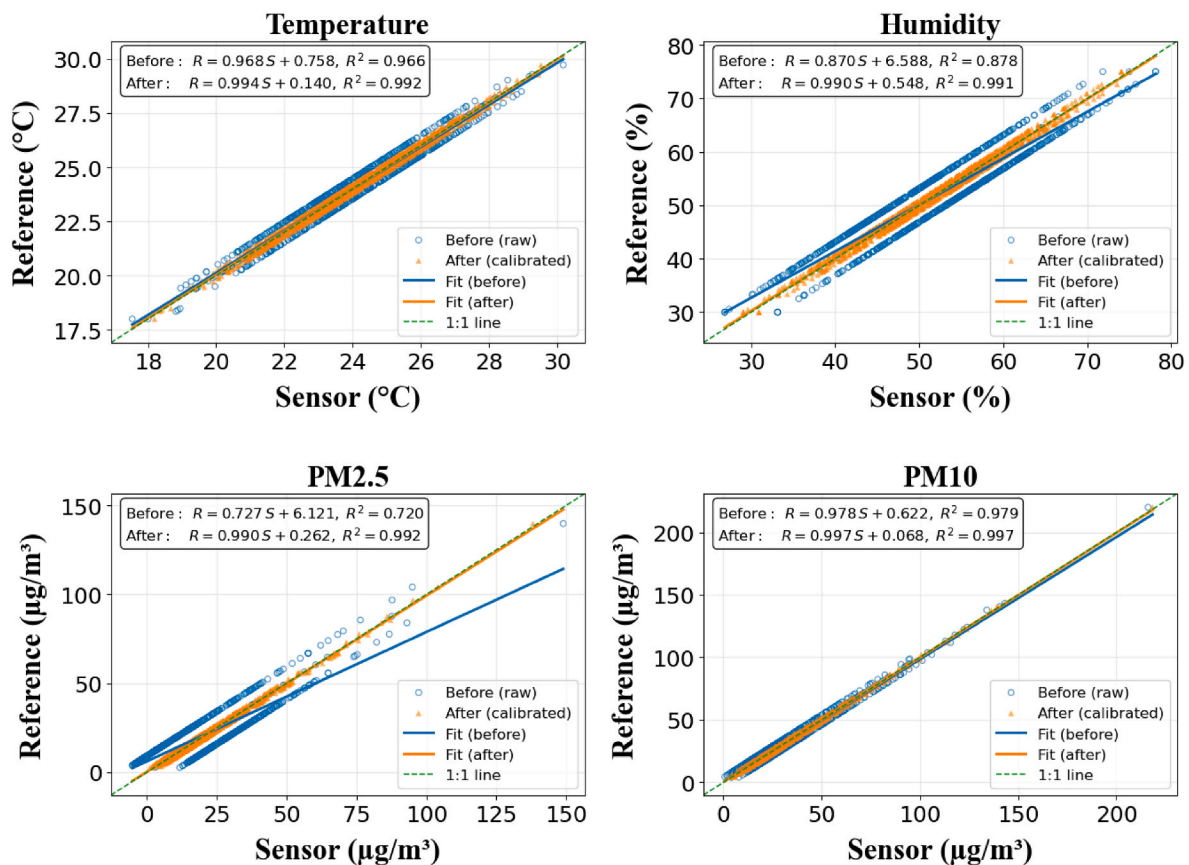


Fig. 3. Scatter correlation of raw and calibrated sensor measurements against reference values: (a) temperature (°C), (b) relative humidity (%), (c) PM_{2.5} (µg/m³), and (d) PM₁₀ (µg/m³), including linear regression fits and the 1:1 reference line.

Table 2
Evaluation metrics before and after calibration.

Parameters	Before calibration			After calibration		
Metrics	RMSE	MAE	MSE	RMSE	MAE	MSE
Temperature (°C)	0.36	0.28	0.13	0.19	0.16	0.03
Relative humidity (%)	3.03	2.91	9.15	0.79	0.67	0.62
PM _{2.5} (µg/m ³)	8.70	8.29	7.62	1.28	0.98	1.65
PM ₁₀ (µg/m ³)	3.05	2.42	9.30	1.19	0.93	1.42

Post-calibration, the sensor demonstrated a substantial improvement in accuracy, with RMSE, MAE, and MSE reduced to 1.28, 0.98, and 1.65, respectively. In addition to reduced error metrics, the post-calibration sensor output exhibited lower noise and variability, showing closer values with the reference measurements. Despite the challenges noted in other studies, where some bypassed calibration in favor of manufacturer specifications [71], this approach can integrate regular recalibration processes to maintain the long-term accuracy of the air quality monitoring system.

3.1.3. Cloud computing

Cloud computing is essential for enabling secure storage, real-time monitoring, and analysis of indoor environmental data. Approximately 65% of existing studies favor cloud-based servers due to their ubiquitous access and flexible integration capabilities. These studies commonly employ structured query language (SQL) databases for managing indoor environmental data repositories. However, a notable trend is emerging toward the adoption of IoT-oriented data storage solutions, with ThingSpeak being identified as the preferred platform in 20% of the reviewed literature [72]. In the present study, ThingSpeak cloud server was utilized as an open API and IoT-focused cloud environment, enabling seamless acquisition and storage of sensor data. ThingSpeak cloud architecture supports the creation of configurable channels, each capable of being set as public or private, which serve as the core structure for organizing and managing data. These channels include multiple fields, allowing the systematic classification of data based on specific indoor environmental parameters. This design enhances accessibility, scalability, and integration with real-time IoT sensor networks.

3.2. Real-time monitoring and prediction protocol

3.2.1. Real-time monitoring protocol

The real-time monitoring system was deployed in the study site described in Section 3.1, using occupant-level placement to capture breathing-zone conditions under routine office operation. The setup was positioned at a height of 1.1 m, within the recommended breathing zone range of 75–120 cm, to ensure accurate environmental measurements [73]. The system utilized a Raspberry Pi 4B to interface with two sensors: the PM2008M for particulate matter (PM_{2.5} and PM₁₀) and the DHT22 for temperature and relative humidity. Each sensor reading was calibrated before deployment. A custom Python script executed within Visual Studio Code handled continuous data acquisition at 1-min intervals.

The data were printed in real-time to the Raspberry Pi terminal and transmitted immediately to designated fields on ThingSpeak channels using individual API keys. Each parameter, PM_{2.5}, PM₁₀, temperature, and relative humidity, was collected, calibrated, and transferred to the cloud for live visualization and further analysis. ThingSpeak IoT-enabled infrastructure facilitated structured storage, retrieval, and presentation of the data, while supporting Excel-based downloads through secure API access. Continuous environmental data were collected using this system over a six-month period with a 1-min sampling interval, yielding 259,200 samples for each sensor. The extended data collection duration was selected to capture seasonal variability, ensure long-term sensor stability, and identify periods of intensive cooling demand to support subsequent phases of the study. Fig. 4 illustrates the real-time

Channel Stats

Created: 4 months ago

Last entry: less than a minute ago

Entries: 504256

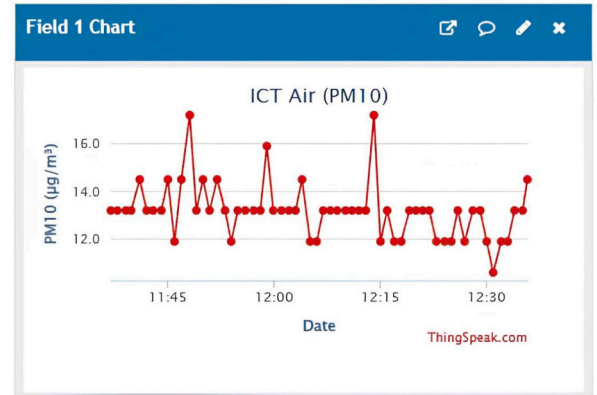


Fig. 4. Real-time PM₁₀ monitoring with 1-min interval updates.

data monitoring interface from the ThingSpeak IoT platform for PM₁₀ readings. The selected channel, titled ICT Air (PM₁₀), displays live data acquired from the PM2008M sensor. The chart within the figure shows fluctuations in PM₁₀ concentration (µg/m³) over time, emphasizing the continuous and responsive behavior of the sensor. This real-time visualization confirms the effective transmission and dynamic cloud-based monitoring capability of the proposed system, enabling immediate observation and retrospective analysis of indoor environmental trends.

3.2.2. Real-time prediction protocol

The Gated Recurrent Unit (GRU) is a streamlined alternative to the Long Short-Term Memory (LSTM) model, offering similar predictive performance with reduced computational complexity. By employing a simplified gating mechanism, GRU models are easier to train and require fewer parameters compared to traditional LSTMs, making them an efficient choice for sequential data modeling. The incorporation of attention mechanisms further improves GRU performance by allowing the model to prioritize relevant portions of the input sequence, strengthening its capability to capture intricate temporal relationships. While DL has increasingly been applied in environmental studies, especially for outdoor air quality forecasting, the use of attention-enhanced GRU models for indoor environmental prediction remains limited. This study explores the application of a proposed GRU model integrated with the simplified attention-mechanism (GRU-SAM) for real-time indoor environmental parameters prediction. The GRU-SAM model was specifically selected for this real-time indoor environmental prediction task due to its optimal balance of computational efficiency and temporal modeling capability. The proposed approach aims to enhance prediction accuracy within a 1-min processing interval while minimizing training loss. Due to its efficient gating structure, the GRU model is well-suited for learning from sequential sensor data with minimal computational overhead. This makes it particularly advantageous for short-term indoor environmental prediction scenarios that demand both rapid inference and the ability to learn from recent data trends.

To validate computational efficiency beyond deep learning models, the proposed GRU-SAM model was additionally compared against traditional machine learning and statistical models, including Random Forest (RF), XGBoost, and AutoRegressive Integrated Moving Average (ARIMA). These models were selected because they are commonly regarded as lighter alternatives for short-term forecasting and are frequently deployed in resource-constrained settings. For comparison,

all baseline models were evaluated under the same real-time prediction protocol, using identical sliding-window lengths, prediction horizons, and chronological train-test partitions as the GRU-SAM model. Time-series inputs for RF and XGBoost were constructed using lagged observations and simple rolling statistics derived from the same historical window available to the GRU-SAM model. Performance metrics and computational characteristics of these classical baselines are reported alongside deep learning models in Section 4 to assess whether the proposed GRU-SAM provides a meaningful trade-off between prediction accuracy and computational efficiency.

As shown in Fig. 5, the GRU is a streamlined variant of the Recurrent Neural Network (RNN) that utilizes gating mechanisms to effectively manage temporal dependencies in sequential data. At every time step t , the model gets an input vector x_t and the latest hidden state h_{t-1} . The GRU cell computes two primary gates: the reset gate r_t , and the update gate z_t . The reset gate controls the degree to which prior hidden state information is retained or removed. The update gate controls how much of the past information is retained for the current step. Using the reset gate output, the model computes the candidate activation $\tilde{h}_t$, which is a potential update to the hidden state derived through a tanh activation. Ultimately, the hidden state is revised by combining the prior state with the candidate state, moderated by the update gate weighting. This gating mechanism enables GRU to efficiently model for short-term dependencies, making it particularly useful for time series prediction applications.

A single-layer GRU architecture integrated with an attention mechanism is employed for short-term prediction. The model receives a fixed-length input sequence representing historical data through an input layer. This sequence is processed by the GRU layer, which outputs hidden states corresponding to each time step. These hidden states are passed to an attention layer that assigns weights to each state based on its relevance to the prediction task. A context vector is then computed as a weighted sum of these hidden states. This context vector is subsequently fed into a fully connected layer, followed by a regression layer to estimate the value for the next interval. This architecture effectively captures temporal dependencies while emphasizing the most informative parts of the input sequence, making it well-suited for the real-time prediction.

The GRU computations are:

1. Reset gate: $r_t = \sigma(W_r \cdot [h_{t-1}, x_t])$
2. Update gate: $z_t = \sigma(W_z \cdot [h_{t-1}, x_t])$
3. Candidate activation: $\tilde{h}_t = \tanh(W_h \cdot [r_t \cdot h_{t-1}, x_t])$
4. Final hidden state: $h_t = (1 - z_t) \cdot h_{t-1} + z_t \cdot \tilde{h}_t$

The temporal weighting mechanism integrated into the proposed GRU-based model adopts a lightweight, parameter-free temporal weighting formulation designed for efficient short-term time-series

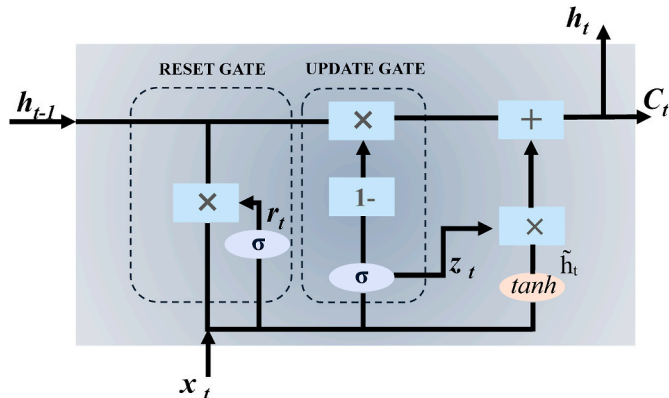


Fig. 5. GRU cell architecture.

prediction. Unlike conventional attention mechanisms such as those proposed by Bahdanau, Luong, or the scaled dot-product attention used in Transformer architectures, the present implementation eliminates trainable compatibility functions to enforce strict parameter efficiency. Instead, a fixed, activation-based temporal scoring rule is employed, where each time step is assigned a proxy score derived from the aggregate magnitude of the GRU hidden-state activations:

$$e_t = \frac{1}{d} \sum_{k=1}^d |X_{t,k}| \quad (1)$$

where $X_t \in \mathbb{R}^d$ denotes the GRU hidden state at time step t , and d represents the dimensionality of the hidden state (i.e., the number of GRU units). The resulting temporal scores are then normalized using the softmax function to obtain parameter-free weighting coefficients:

$$\alpha_t = \text{softmax}(e_t) \quad (2)$$

Finally, the context vector is computed as a weighted sum of the hidden states:

$$Z = \sum_t \alpha_t \cdot X_t \quad (3)$$

This formulation provides a computationally efficient alternative to learn attention mechanisms by biasing temporal aggregation toward time steps exhibiting higher aggregate activation, while preserving stability and suitability for real-time, resource-constrained deployment.

Let $h_t \in \mathbb{R}^d$ denote the GRU hidden state at time step t . In the proposed lightweight formulation, the temporal score is computed from the aggregate magnitude of hidden activations, which corresponds to a fixed symmetric projection of h_t . Under a linear readout or local linear approximation of the predictor, the contribution of a given time step to the output can be expressed as a weighted inner product $w^T h_t$. In the absence of prior preference over hidden dimensions and to enforce strict parameter efficiency, choosing a constant vector $w \times 1$ yields a parameter-free proxy score proportional to the aggregate activation of h_t . While this formulation does not learn an adaptive alignment function, it provides a computationally efficient and stable temporal weighting mechanism aligned with the deployment-oriented objective of model lightweighting.

After computing the proxy score, the softmax function yields normalized activation-based proxy weights for each time step. The resulting context vector is a weighted combination of hidden states, which biases temporal aggregation toward time steps exhibiting higher aggregate activation under this fixed scoring rule. Attention is used in the sense of parameter-free temporal weighting; the weights should be interpreted as a lightweight saliency proxy rather than a learned importance measure. Fig. 6 shows the proposed architecture of GRU-SAM model.

During closed-loop streaming, occasional missing samples and rare outliers were observed, primarily due to Wi-Fi interruptions, HTTP timeout events, and sporadic sensor errors. To preserve the continuity required for 1-min sliding-window inputs and uninterrupted real-time inference, missing timestamps were imputed using linear interpolation between adjacent samples. Specifically, for a missing observation at time t , located between two valid samples at times t_1 and t_2 , with corresponding values $x(t_1)$ and $x(t_2)$, the interpolated value $x(t)$ was computed as

$$x(t) = x(t_1) + \frac{t - t_1}{t_2 - t_1} [x(t_2) - x(t_1)] \quad (4)$$

which provides a conservative estimate for short communication gaps [74,75]. Outliers were screened using a Tukey boxplot (interquartile range, IQR) criterion, where the IQR was defined as the difference between the third and first quartiles ($Q_3 - Q_1$). Data points falling below $Q_1 - 1.5 \times \text{IQR}$ or above $Q_3 + 1.5 \times \text{IQR}$ were flagged as outliers. Flagged

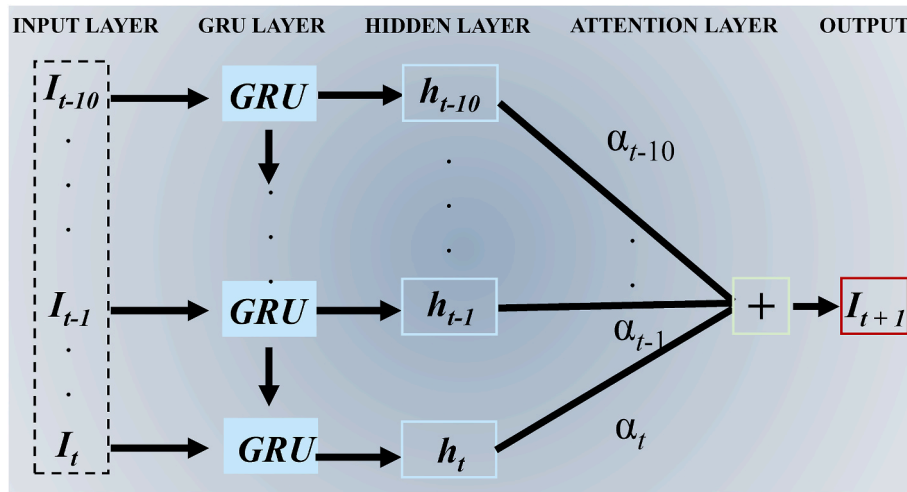


Fig. 6. GRU-SAM model architecture, consisting of GRU layers, hidden layers, and an attention layer for output prediction.

outliers were then removed and replaced using linear interpolation to prevent abnormal values from propagating into the prediction pipeline and to maintain a continuous sequence for downstream inference [76]. To avoid data leakage in time-series forecasting, the dataset used for model training and evaluation was partitioned in chronological order. Earlier observations were used for training, while the most recent observations were reserved for testing. Sliding-window sequences were generated independently within each data split, ensuring that no test input window contained future information relative to the training data.

To enable real-time prediction, the proposed system integrates the GRU-SAM model with the ThingSpeak IoT platform, allowing for continuous data acquisition and dynamic forecast updates. The GRU-SAM model, implemented in Python (version 3.11.2), uses ThingSpeak API to retrieve real-time indoor environmental data from designated channel fields and writes the predicted values back to a separate prediction channel field. Every minute, the edge platform (Raspberry Pi) acquires the latest measurements, processes the most recent time window of data, and generates a short-term forecast for the corresponding air quality parameter. The final forecast value is then transmitted to the prediction channel field, ensuring its immediate availability for proactive control signal generation. This design ensures that predictions are continuously updated within a 1-min computational cycle, with older data discarded as new data streams into the monitoring channel. Operating in a continuous loop, the model autonomously updates predictions and sends them to the ThingSpeak platform. This configuration ensures that the system remains adaptive and responsive, supporting timely and informed decision-making in indoor environments.

To quantify real-time performance, system latency and streaming inference availability were evaluated during continuous deployment. Each operational cycle was defined from the timestamp of sensor data acquisition to the successful publication of the corresponding predicted values, encompassing data acquisition, prediction pipeline execution, and result publishing. A missing inference event was recorded when no valid prediction output was produced within the scheduled cycle due to transient communication or sensing disruptions. For subsequent analysis, missing values in the latency time series were addressed using linear interpolation, while abnormal latency values were identified and constrained using a boxplot-based outlier detection approach.

4. Results and discussion

In this section, the performance of the proposed ICTAir framework and the GRU-SAM prediction model is evaluated through four complementary analyses. First, the statistical volatility and temporal dynamics of the monitored indoor variables (temperature, relative humidity,

PM_{2.5}, and PM₁₀) are quantified to characterize variability patterns and justify the need for robust prediction under real-world fluctuations. Second, the predictive accuracy of GRU-SAM is systematically benchmarked against multiple baseline models across all four parameters using standard error metrics. Third, a training-testing window optimization study is performed to identify the temporal configuration that yields the most reliable forecasting performance. Finally, real-time deployability is validated by analyzing end-to-end system latency, streaming inference reliability, and the inference-time and memory-footprint characteristics of the evaluated models.

4.1. Statistical volatility and temporal dynamics of indoor environmental variables

To assess the intrinsic difficulty of short-horizon prediction, the volatility and temporal dynamics of the monitored indoor variables were quantitatively evaluated using 1-min resolution data collected over a representative one-week period. Key indicators include the coefficient of variation (CV), mean absolute first difference (MAD Δ), 95th-percentile absolute change (P95 ($|\Delta|$)), skewness, kurtosis, and autocorrelation at 15- and 60-min lags. Table 3 summarizes indicators of variability, abrupt change, and temporal persistence, which directly characterize the non-stationary and event-driven nature of the indoor environment.

Temperature and relative humidity exhibit relatively smooth behavior, with low coefficients of variation (8.49% and 14.01%, respectively) and small mean absolute changes (MAD $\Delta \leq 0.13$), reflecting comparatively stable thermal dynamics over short horizons. In contrast, particulate matter concentrations show markedly higher variability, with PM_{2.5} and PM₁₀ exhibiting coefficients of variation exceeding 60% and large 95th-percentile absolute changes (1.75 and 6.60 $\mu\text{g}/\text{m}^3$, respectively), indicating frequent abrupt shocks. The positive skewness (3.35 for PM_{2.5} and 2.35 for PM₁₀) and high kurtosis values (28.51 and 7.47) confirm the spike-dominated nature of particulate matter dynamics. Autocorrelation is strong at 15 min for all variables and remains moderate-to-strong at 60 min, while PM exhibits comparatively lower persistence than thermal variables, reflecting its sensitivity to transient indoor events.

These statistical characteristics highlight the non-stationary behavior of indoor air pollutants, representing the inherent difficulty of short-horizon prediction and motivating the sequence-based deep learning models evaluated in the subsequent section. Fig. 7 illustrates the weekly evolution of the thermal variables and serves as a visual counterpart to the low-volatile statistics reported in Table 3.

Both temperature (Fig. 7a) and relative humidity (Fig. 7b) show predominantly smooth trajectories with gradual rises and declines

Table 3
Statistical volatility and temporal dynamics of indoor environmental variables.

Parameters	CV (%)	MAD (Δ)	P95 ($ \Delta $)	Skewness	Kurtosis	ACF (15 min)	ACF (60 min)
T ($^{\circ}\text{C}$)	8.49	0.01	0.07	0.36	1.46	0.99	0.90
RH (%)	14.01	0.13	0.44	0.06	1.16	0.98	0.88
PM _{2.5} ($\mu\text{g}/\text{m}^3$)	62.28	0.62	1.75	3.35	28.51	0.83	0.77
PM ₁₀ ($\mu\text{g}/\text{m}^3$)	57.25	1.76	6.60	2.35	7.47	0.88	0.74

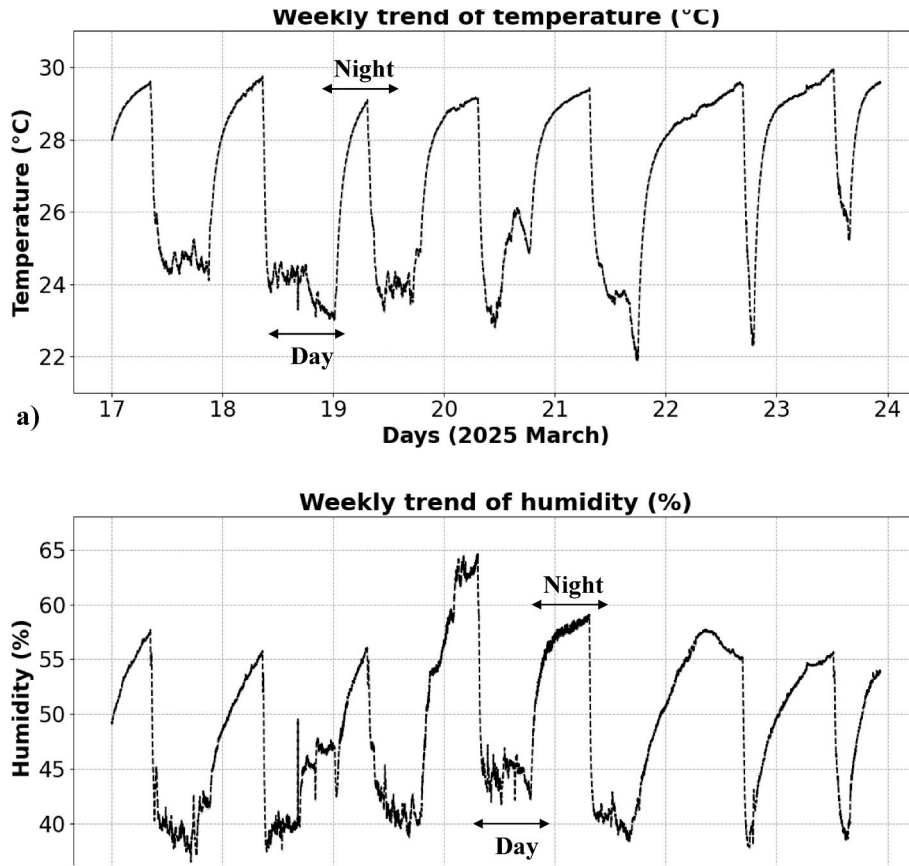


Fig. 7. High-resolution weekly trends: (a) temperature ($^{\circ}\text{C}$); (b) relative humidity (%).

rather than abrupt discontinuities, consistent with low short-term variability and strong temporal persistence. The temperature profile repeatedly increases toward the upper range (29–30 $^{\circ}\text{C}$) and then transitions to lower levels (23–25 $^{\circ}\text{C}$), indicating repeated regime shifts within the week; however, these shifts remain comparatively structured and time-correlated relative to particulate variables. Relative humidity follows a complementary pattern (38–65%), with elevated plateaus and slower recoveries, reflecting moisture retention and lagged humidity responses under varying indoor conditions.

As shown in Fig. 8, the non-stationary behavior of particulate matter concentrations, in contrast to the smoother thermal variables. Both PM_{2.5} (Fig. 8a) and PM₁₀ (Fig. 8b) exhibit abrupt spikes followed by rapid decay, consistent with the elevated MAD Δ and P95 ($|\Delta|$) values, as well as pronounced positive skewness and kurtosis reported in Table 3. PM_{2.5} concentrations intermittently exceed 35 $\mu\text{g}/\text{m}^3$, while PM₁₀ peaks reach above 50 $\mu\text{g}/\text{m}^3$, indicating short-lived but intense excursions. These heavy-tailed distributions and burst-like transitions reflect event-driven dynamics and non-stationary behavior, which substantially increases the difficulty of accurate short-horizon prediction. At the minute resolution, successive sharp rises and collapses occur over time scales of only a few minutes, emphasizing the limited temporal persistence observed in the lower autocorrelation values relative to thermal variables. These statistical results provide quantitative visual evidence that

particulate matter prediction requires models capable of rapidly adapting to transient, high-impact events, motivating the use of attention-enhanced sequence learning.

4.2. Accuracy analysis of GRU-SAM with baseline models

To comprehensively evaluate the predictive capability of the proposed GRU-SAM model, a broad range of baseline models was considered for accuracy analysis. In addition to established deep learning (DL) architectures commonly applied to short-term time-series forecasting namely LSTM, LSTM-SAM, BiLSTM, BiLSTM-SAM, and GRU, the comparison was extended to include traditional machine learning and statistical time-series models. Specifically, Random Forest (RF), Extreme Gradient Boosting (XGBoost), and the AutoRegressive Integrated Moving Average (ARIMA) model were incorporated. All DL models were implemented using a single-layer architecture to ensure consistency in computational complexity and training efficiency. The simplified attention mechanism (SAM) was applied uniformly across compatible DL variants to isolate its contribution to predictive accuracy. For comparison, identical training–testing splits, input window lengths, and evaluation metrics were used across all models. The ARIMA model was configured using standard order selection based on autocorrelation and partial autocorrelation analysis, while RF and XGBoost models were

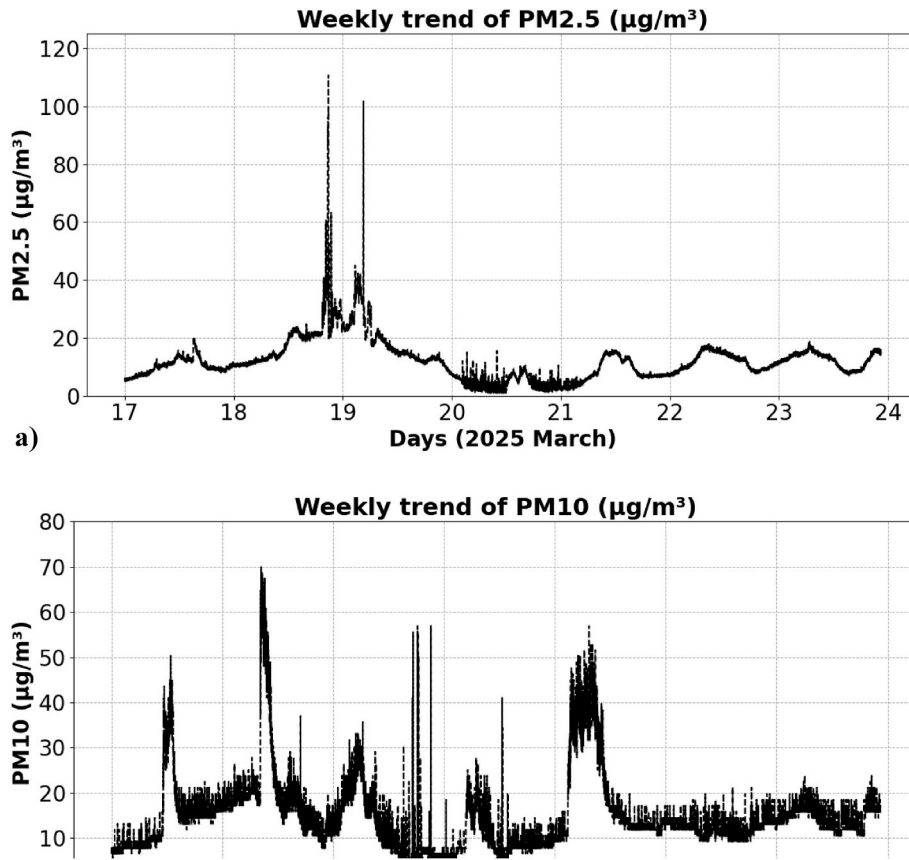


Fig. 8. High-resolution weekly concentration trends: (a) PM_{2.5} (µg/m³); (b) PM₁₀ (µg/m³).

tuned using conservative hyperparameter settings to avoid overfitting and to reflect practical lightweight deployment scenarios.

During the comparative analysis, several key hyperparameters were standardized across all models to ensure consistent evaluation and fair benchmarking. The input sequence length was set to 15 time steps based on Bayesian optimization combined with short-horizon sensitivity analysis, which indicated that this window sufficiently captures the dominant short-term temporal dynamics of the monitored indoor environmental variables while avoiding unnecessary computational overhead. The number of hidden units was fixed at 500 across all deep learning models to reduce the risk of under-parameterization and to maintain comparable representational capacity among the evaluated architectures during benchmarking. Although a hidden size of 500 may appear large for a lightweight edge model, the resulting parameter count remains on the order of 10^6 , corresponding to only a few megabytes of model weights (excluding runtime framework overhead) in single-precision floating-point format. Given the available memory and computational resources of the Raspberry Pi 4B, this configuration remains feasible for comparative accuracy analysis, while smaller configurations (e.g., 256 units) can be preferred for resource-constrained deployment. The training process employed the Adam optimizer with a learning rate of 0.001, a mini-batch size of 64, and a maximum of 100 epochs to enable stable convergence. To mitigate the risk of exploding gradients, gradient clipping with a threshold of 1 was employed. For data preparation, the model used the most recent 180 time-series data points, with a 90:10 train-test split to simulate real-time forecasting scenarios. These hyperparameters were selected through the process of Bayesian optimization methods to balance computational efficiency and predictive accuracy. The average runtime ranged from 10 s to 2 min for each model and corresponds to batch-level evaluation during testing, supporting suitability for real-time prediction.

To rigorously assess the predictive performance of the proposed

models forecasting, three widely accepted evaluation metrics were employed: RMSE, MAE, and MSE. These metrics provide a comprehensive evaluation by quantifying different aspects of prediction error, as outlined below.

MAE: MAE quantifies the average absolute difference between the predicted and actual values. It is expressed as:

$$MAE = (1/N) \sum |y_i - \hat{y}_i| \quad (5)$$

where y_i and $\hat{y}_i$ denote the actual and predicted values, respectively, and N is the total number of observations. A smaller MAE reflects higher prediction accuracy.

MSE: MSE calculates the average of the squared differences between predicted and actual values:

$$MSE = (1/N) \sum (y_i - \hat{y}_i)^2 \quad (6)$$

Unlike MAE, MSE gives more weight to larger errors, making it more sensitive to outliers and extreme deviations.

RMSE: RMSE is derived as the square root of MSE. It brings the error metric back to the original unit of the target variable, enhancing its interpretability:

$$RMSE = \sqrt{(1/N) \sum (y_i - \hat{y}_i)^2} \quad (7)$$

Due to its sensitivity to large errors, RMSE is especially useful for identifying performance degradation in volatile or anomalous data regions. Together, these metrics offer a well-rounded evaluation framework to compare the accuracy and robustness of different prediction models. The performance analysis categorized into two main groups: (1) thermal comfort parameters (temperature, relative humidity), and (2) particulate matter (PM_{2.5} and PM₁₀) concentrations.

4.2.1. Thermal comfort parameters

As illustrated in Fig. 9a, GRU-SAM achieves the lowest prediction errors for temperature, with MSE = 0.01, RMSE = 0.07, and MAE = 0.06. The base GRU model also performs strongly (MSE = 0.27, RMSE = 0.52, MAE = 0.51), confirming the suitability of GRU-based architectures for modeling smoothly varying and autocorrelated temperature dynamics. In contrast, LSTM and LSTM-SAM exhibit substantially higher errors, indicating that their more complex memory structures provide limited benefit for this signal and may introduce unnecessary training overhead. Among the non-DL baselines, ARIMA demonstrates relatively competitive performance for temperature (MSE = 0.075, RMSE = 0.095, MAE = 0.112), reflecting the quasi-stationary and linear characteristics of short-term indoor temperature variations. RF and XGBoost further lag behind, with XGBoost showing notably higher MSE (0.332) and RMSE (0.542), indicating reduced robustness to temporal dependencies.

A similar performance hierarchy is observed for relative humidity prediction (Fig. 9b). GRU-SAM outperforms all competing models (MSE = 2.80, RMSE = 1.67, MAE = 1.55), followed by ARIMA (MSE = 2.872, RMSE = 1.748, MAE = 1.640). Although ARIMA yields acceptable accuracy, it is unable to capture the nonlinear and externally influenced fluctuations characteristic of indoor humidity dynamics. RF and XGBoost exhibit larger errors, particularly in MSE, suggesting sensitivity to noise and limited generalization under rapidly changing humidity conditions. These results indicate that integrating a lightweight temporal weighting mechanism within the GRU framework enables selective emphasis on recent and dynamically relevant observations, allowing GRU-SAM to adapt more effectively to subtle

temporal shifts and external perturbations than non-DL baselines that rely on fixed linear assumptions or static feature aggregation.

4.2.2. Particulate matter concentrations

For PM_{2.5} prediction (Fig. 10a), GRU-SAM achieves the lowest error values (MSE = 0.36, RMSE = 0.60, MAE = 0.46), followed by the base GRU model (MSE = 0.90, RMSE = 0.95, MAE = 0.91). Traditional statistical and machine learning models exhibit lower accuracy: ARIMA yields MSE = 0.965 and RMSE = 1.225, while RF and XGBoost show even higher error levels. These results indicate that PM_{2.5} dynamics exhibit nonlinear temporal patterns and abrupt fluctuations that cannot be adequately modeled using linear autoregressive assumptions or static ensemble learners. A comparable trend is observed for PM₁₀ forecasting (Fig. 10b). GRU-SAM demonstrates higher accuracy (MSE = 0.84, RMSE = 0.91, MAE = 0.77) than GRU, BiLSTM-based models, and other non-DL baselines. ARIMA and RF exhibit moderate performance, while XGBoost records the higher errors (MSE = 3.822). The relatively poor performance of bidirectional architecture suggests that backward temporal context may introduce unnecessary complexity or noise for particulate matter prediction.

Although ARIMA demonstrates reasonable accuracy for temperature and relative humidity, its performance degrades for particulate matter prediction, where nonlinear dynamics and transient spikes dominate. These results demonstrate that, although classical statistical models may be adequate for specific stable signals, a lightweight GRU with temporal weighting achieves consistent performance across heterogeneous indoor parameter-level prediction. During the analysis, it was observed that the GRU-SAM model consistently achieved better predictive performance

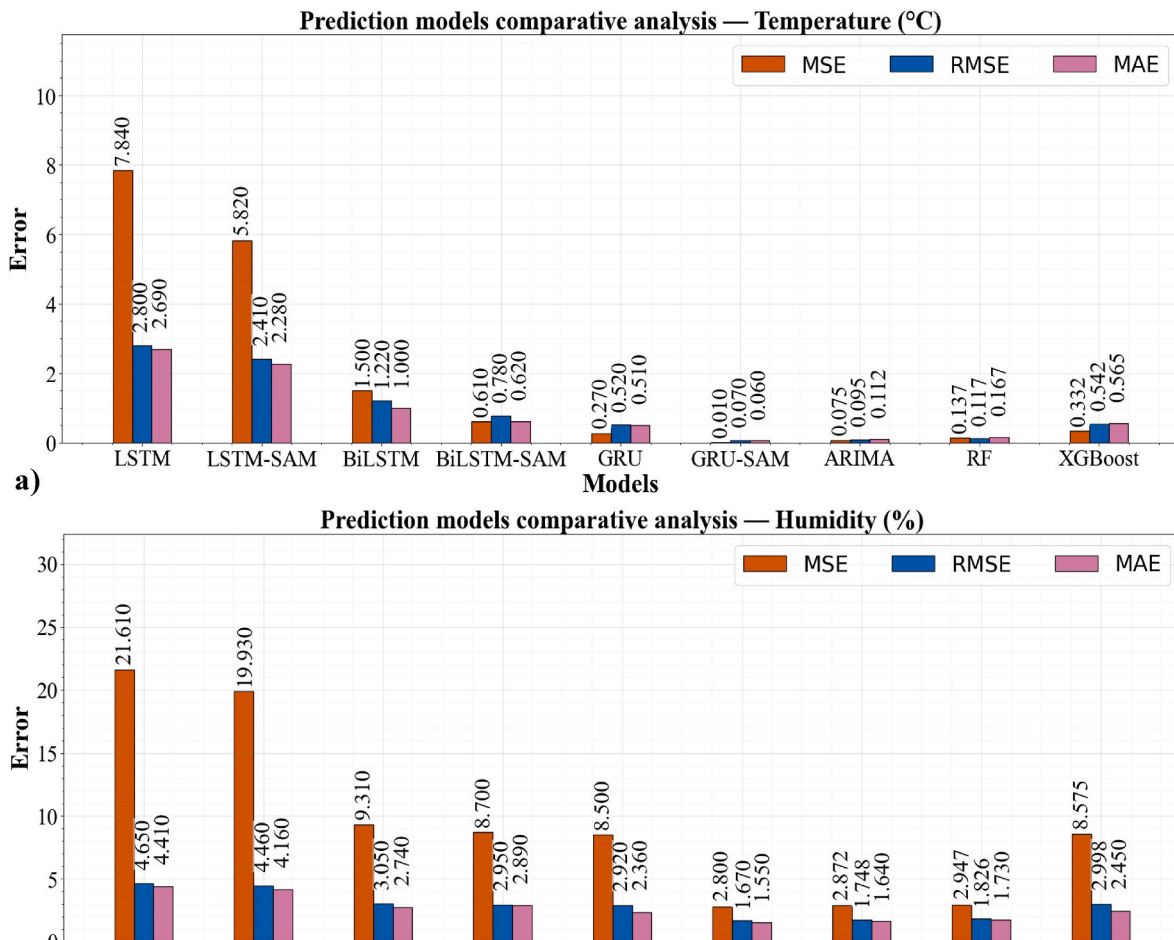


Fig. 9. Comparative performance evaluation of deep learning, statistical, and machine learning models for thermal comfort parameters: (a) temperature (°C); (b) relative humidity (%).

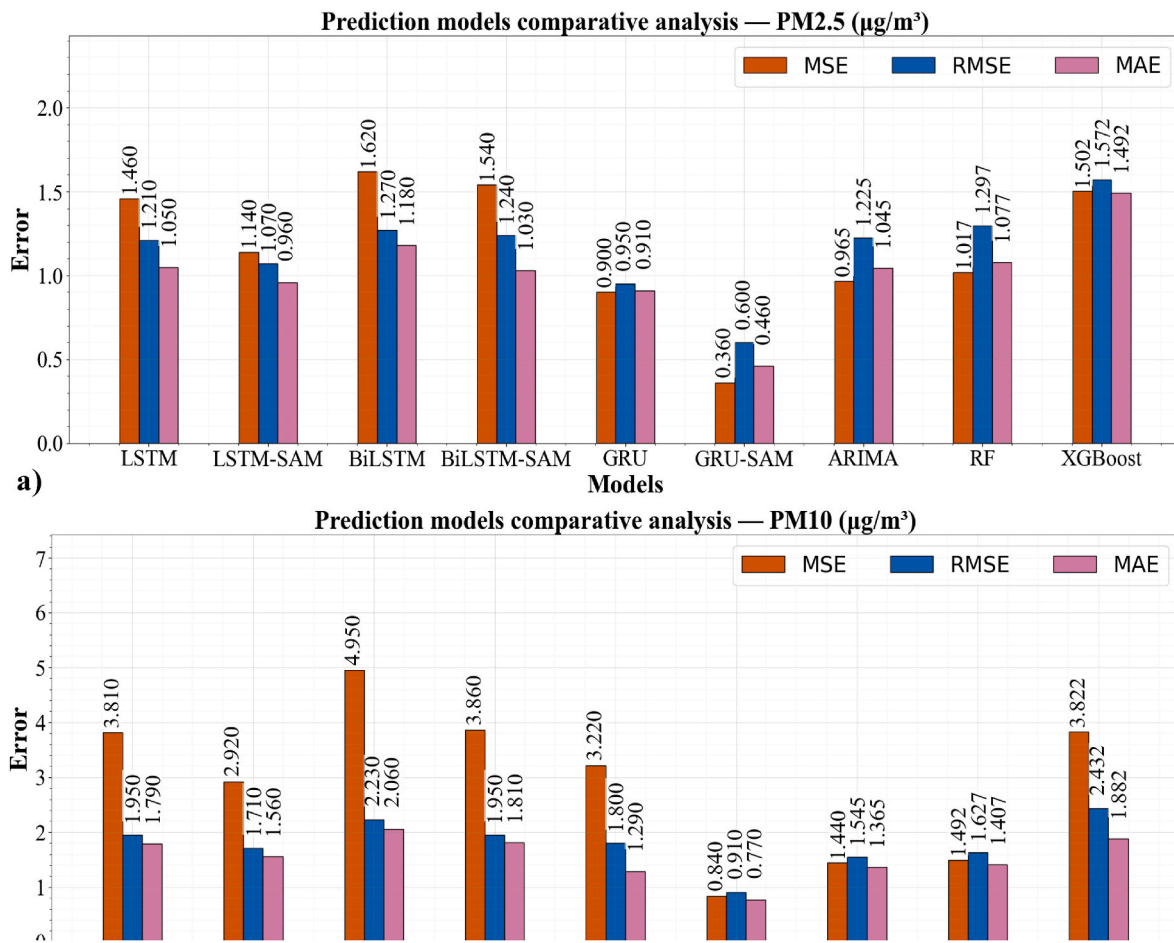


Fig. 10. Comparative performance evaluation of deep learning, statistical, and machine learning models for particulate matter concentrations: (a) PM_{2.5} (µg/m³); (b) PM₁₀ (µg/m³).

across all parameters, but a closer inspection reveals that the lowest errors were obtained for temperature and particulate matter (PM_{2.5} and PM₁₀), with relatively higher errors observed in relative humidity prediction. Temperature exhibited the most stable and predictable behavior, enabling the model to achieve near-zero errors (RMSE = 0.07). Interestingly, particulate matter predictions also showed relatively low error values (e.g., PM_{2.5} with RMSE = 0.60; PM₁₀ with RMSE = 0.91), despite their well-known volatility.

In contrast, relative humidity forecasts exhibited higher error levels (RMSE = 1.67). Although short-term ventilation and infiltration events can induce abrupt moisture variations, this discrepancy should be interpreted considering the thermodynamic coupling between temperature and relative humidity. Relative humidity is inherently more sensitive to transient moisture sources and is also affected by sensor-specific characteristics, as RH sensors typically exhibit slower dynamic response and hysteresis compared with temperature sensors, which can increase apparent prediction error during rapid changes. In this study, relative humidity was modeled using a univariate temporal formulation based solely on historical RH measurements, without incorporating temperature as an exogenous predictor. This parameter-level modeling strategy was adopted to maintain a consistent prediction framework across variables; however, incorporating temperature within a multivariate formulation is expected to further improve relative humidity prediction accuracy and will be investigated in future work.

4.3. Performance optimization of GRU-SAM model by training and testing data windows

A comprehensive analysis was conducted to optimize the configuration of training and prediction windows with the goal of maximizing prediction accuracy across all four parameters. Training window lengths varied from 1 to 3 h of the most recent historical data, while prediction horizons ranged from 5 to 15 min into the future. The selection of training window lengths was strategically aimed at minimizing computation time to ensure that system latency remained under 1 min. Meanwhile, the prediction windows were defined based on the expected response time of indoor environmental control systems such as air conditioners for thermal comfort parameters (temperature, relative humidity) and True HEPA filters for particulate matter (PM_{2.5} and PM₁₀) concentrations, required to bring the indoor air parameter concentration under threshold values in indoor conditions. This experimental design yielded nine unique training-prediction combinations, establishing a robust and adaptable framework for short-term forecasting in smart indoor environments. All data used in this study were retrieved from the ThingSpeak cloud platform and segmented into intervals for each parameter to ensure consistent granularity and alignment across input sequences.

Table 4 illustrates the prediction performance of the GRU-SAM model for four key indoor environmental indicators. The evaluation includes a total of nine training-prediction combinations, consisting of three training durations (1, 2, and 3 h) and three prediction windows (5, 10, and 15 min). These scenarios are analyzed to identify the optimal configuration for short-term prediction. Performance assessment is

Table 4

Performance optimization metrics of GRU-SAM model with training and prediction window.

Parameter	Metric	1–5 (h-m)	1–10 (h-m)	1–15 (h-m)	2–5 (h-m)	2–10 (h-m)	2–15 (h-m)	3–5 (h-m)	3–10 (h-m)	3–15 (h-m)
Temperature	RMSE	0.22	0.69	0.74	0.96	1.47	1.93	1.10	1.65	2.01
	MAE	0.20	0.58	0.67	0.88	1.39	1.83	1.02	1.58	1.87
	MSE	0.05	0.48	0.54	0.93	2.17	3.71	1.21	2.72	4.04
Relative Humidity	RMSE	0.24	1.26	1.95	1.46	1.59	2.02	1.81	2.01	2.30
	MAE	0.24	1.17	1.64	1.40	1.47	1.79	1.76	1.97	2.15
	MSE	0.06	1.59	3.79	2.12	2.52	4.07	3.26	4.06	5.31
PM _{2.5}	RMSE	0.15	0.36	0.47	0.22	0.63	0.60	0.66	1.01	0.94
	MAE	0.12	0.27	0.37	0.22	0.53	0.49	0.42	0.74	0.76
	MSE	0.02	0.13	0.22	0.05	0.44	0.36	0.44	1.02	0.89
PM ₁₀	RMSE	0.32	1.07	1.45	0.60	1.18	0.84	0.69	1.97	1.02
	MAE	0.30	0.95	1.27	0.48	1.03	0.68	0.58	1.74	0.80
	MSE	0.10	1.15	2.10	0.36	1.39	0.71	0.47	3.89	1.03

carried out using three common evaluation metrics: MSE, RMSE and MAE.

For temperature, the GRU-SAM model exhibited varying degrees of accuracy depending on the configuration of the training and prediction windows. The highest accuracy was achieved using a 1-h training window with a 5-min prediction horizon, resulting in the lowest error values across all evaluation metrics, RMSE = 0.22, MAE = 0.20, and MSE = 0.05. As the prediction window extended to 10 and 15 min, the performance declined moderately, with RMSE increasing to 0.69 and 0.74, and MSE rising to 0.48 and 0.54, respectively. A consistent increase in prediction error was also observed as the training window lengthened from 1 to 3 h. For instance, when using a 3-h training window with a 15-min prediction horizon, RMSE and MSE escalated to 2.01 and 4.04, respectively. These results indicate that shorter training windows incorporating more recent data enhance the model responsiveness and precision in short-term temperature prediction.

The prediction of relative humidity exhibited a similar trend to temperature but with greater sensitivity to temporal parameters. Notably, the best performance was achieved with a short training window of 1 h and a 5-min prediction horizon (RMSE = 0.24). However, performance deteriorated significantly with longer prediction horizons and extended training windows, reaching the highest error at 3-h training and 15-min prediction (RMSE = 2.30, MSE = 5.31). This performance degradation reflects the higher short-term volatility of relative humidity, which is highly sensitive to rapid environmental changes. These short-term fluctuations reduce the effectiveness of learning from long-term patterns, suggesting that shorter, more recent historical data is more suitable for accurate relative humidity prediction.

The PM_{2.5} prediction model demonstrated a distinct pattern compared to temperature and relative humidity. While a shorter training window of 1 h with a 5-min prediction yielded the lowest RMSE of 0.15 and the performance did not uniformly degrade with longer temporal parameters. Notably, when extending both the training window to 2 or 3 h and the prediction window to 15 min, the model still maintained relatively low error values, for instance, RMSE = 0.60 (2hrs - 15min) and RMSE = 0.94 (3hrs - 15min), which were even lower than some shorter-prediction counterparts like RMSE = 0.63 (2hrs - 10min) and RMSE = 1.01 (3hrs - 10min). These results suggest that the PM_{2.5} model benefits from longer historical input and extended forecasting horizons due to more stable temporal patterns in fine particulate matter concentrations.

A similar trend to PM_{2.5} was observed in the PM₁₀ prediction results. The PM₁₀ forecasting model demonstrated robustness even under extended temporal settings. The optimum performance was achieved with a 1-h training and 5-min prediction window (RMSE = 0.32). Unlike temperature or relative humidity, the prediction performance for PM₁₀ did not deteriorate significantly with longer input durations. For instance, with a 3-h training window and a 15-min forecast horizon, the model still maintained a low RMSE of 1.02, comparable to shorter settings such as RMSE = 1.18 (2hrs-10min) and RMSE = 1.97 (3hrs-10min). This stable error trend suggests that PM₁₀ prediction also

benefits from a broader historical context and extended forecast horizons due to slower variation and more persistent patterns in particulate matter concentrations at the experimental site.

The comparative evaluation of GRU-SAM performance across various training and prediction window configurations reveals that shorter, more recent training windows combined with shorter prediction horizons consistently produce the most accurate forecasts. The optimal setting was found to be a 1-h training window with a 5-min prediction horizon, which yielded the optimum accuracy across all environmental parameters. These findings highlight the importance of aligning temporal input settings with the unique dynamics of each parameter. For instance, while temperature and relative humidity exhibited clear sensitivity to extended historical inputs, PM_{2.5} and PM₁₀ predictions remained relatively stable, even with longer training and forecasting durations, suggesting that particulate matter trends benefit from broader temporal context.

In real-time indoor environmental monitoring, prediction, and control, such optimization is crucial. The size of the indoor space also influences prediction-response timing; larger spaces require longer durations for control systems to adjust parameter levels within target thresholds, while smaller spaces can respond more rapidly. Even within a 1-h window, indoor environmental parameters exhibit complex and nonlinear intra-window temporal dynamics, including delayed pollutant decay, thermal inertia, and abrupt changes driven by occupancy and ventilation events. These dynamics cannot be adequately captured by shallow models such as MLP or SVR, which treat inputs as independent static samples (C. Xu., 2018 [77]). In contrast, the proposed GRU-SAM model leverages gated recurrent units with attention-enhanced mechanism to automatically learn, retain, and weight informative temporal states while suppressing irrelevant or noisy historical information, enabling robust short-term prediction under rapidly changing indoor conditions. Moreover, failure to optimize these time settings may result in delayed or inefficient system responses, undermining the effectiveness of proactive control. Therefore, prior optimization of training and prediction windows is essential before implementing prediction-driven control strategies, as demonstrated in this study.

4.4. Real-time system latency and streaming inference performance

4.4.1. System latency

The real-time performance of the proposed monitoring and prediction framework was evaluated by analyzing the end-to-end system latency over continuous streaming operation. System latency was defined as the elapsed time between sensor data acquisition and the successful publication of predicted values, encompassing data acquisition, prediction pipeline execution, and results stored within each operational cycle. Fig. 11a presents the distribution of end-to-end system latency measured across 1000 consecutive acquisition-prediction cycles. The latency exhibits a unimodal and stable distribution with a mean of 41.9 s and a median of 41.6 s. All cycles were completed within the predefined 60 s real-time constraint, demonstrating 100% system reliability under

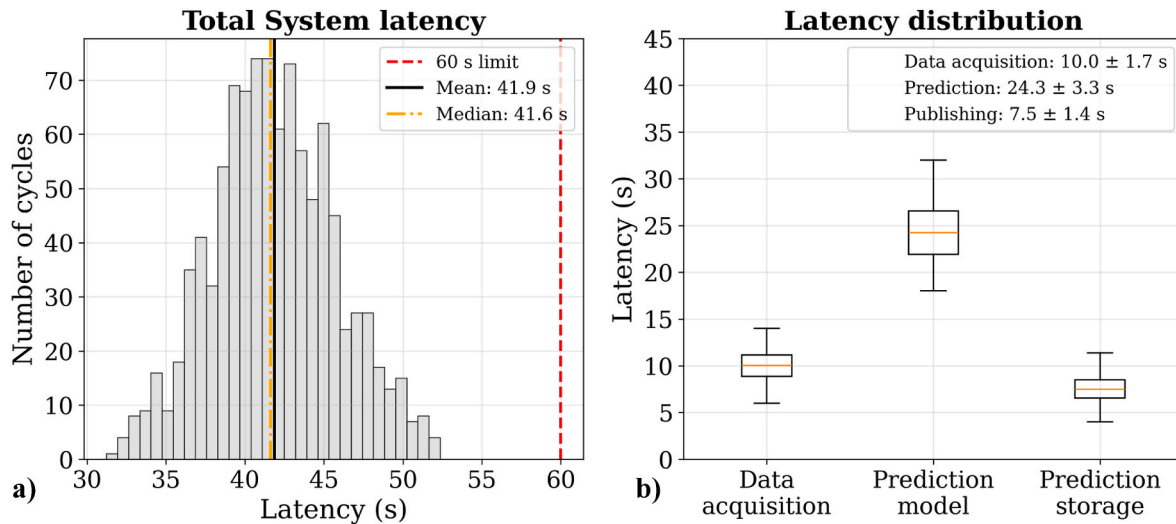


Fig. 11. Real-time system latency characteristics of the proposed framework: (a) End-to-end latency distribution over 1000 continuous acquisition–prediction cycles; (b) Stage-wise latency distribution for data acquisition, prediction model, and prediction storage.

continuous operation. The margin between the observed latency distribution and the 60 s limit confirms that the proposed framework can sustain uninterrupted real-time prediction.

To further interpret the sources of latency, Fig. 11b shows the stage-wise latency breakdown. Data acquisition, which includes sensor sampling and transmission to the cloud platform, required an average of 10.0 ± 1.7 s per cycle. The prediction pipeline constituted the dominant portion of the total latency, with an average execution time of 24.3 ± 3.3 s. This stage encompasses data retrieval, data window construction, and GRU-SAM forward inference on the Raspberry Pi-based edge platform. Storage of predicted values to the cloud required 7.5 ± 1.4 s.

4.4.2. Streaming inference reliability

In addition to satisfying real-time latency constraints, the reliability of streaming inference was evaluated by examining the continuity of prediction availability across consecutive acquisition cycles. Streaming inference reliability was defined as the ability of the proposed framework to generate a valid prediction output for each sensing cycle without interruption. Fig. 12 illustrates the inference availability of 1000 consecutive cycles.

The system achieved an inference reliability rate of 97.7%, indicating that predictions were generated for the vast majority of cycles. Occasional missing inference events, defined as cycles in which a valid prediction output could not be produced within the scheduled interval, were observed, primarily due to transient communication interruptions. To improve robustness under real-world IoT streaming conditions, short sensor data gaps caused by Wi-Fi interruptions and sensor read errors were imputed using linear interpolation between adjacent valid samples. Rare sensor-value outliers were screened using boxplot criterion and replaced using local interpolation to prevent abnormal spikes from

propagating into the streaming inference pipeline. These data-quality controls helped maintain stable prediction availability despite sporadic communication disruptions.

4.4.3. Model inference-time and memory-footprint comparison

To evaluate model efficiencies beyond prediction accuracy, a comparative evaluation of inference latency and memory usage was conducted on an edge computing platform (Raspberry Pi 4B) under a closed-loop streaming inference setup. All models were evaluated using a sliding-window input (window length = 60, batch size = 1), consistent with the real-time prediction framework. Inference latency was measured as the time required to generate a single prediction window, reported as both the mean latency and the 95th-percentile latency (p95) over repeated runs, while memory usage was quantified as the peak resident memory (RAM) observed during inference execution. Table 5

Table 5
Comparison of inference latency and peak memory usage on Raspberry Pi 4B

Models	Inference latency (ms)	Inference latency (ms, p95)	Peak RAM (MB)
Deep learning models			
GRU	778.39	782.51	678.04
GRU-SAM	750.99	716.81	605.34
LSTM	894.61	799.72	614.22
LSTM-SAM	813.79	743.54	618.84
BiLSTM	987.37	996.31	626.53
BiLSTM-SAM	1013.96	1022.53	635.03
Traditional machine learning and statistical models			
ARIMA	153.96	114.22	639.72
Random Forest (RF)	215.08	135.7	642.49
XGBoost	91.27	77.33	658.45

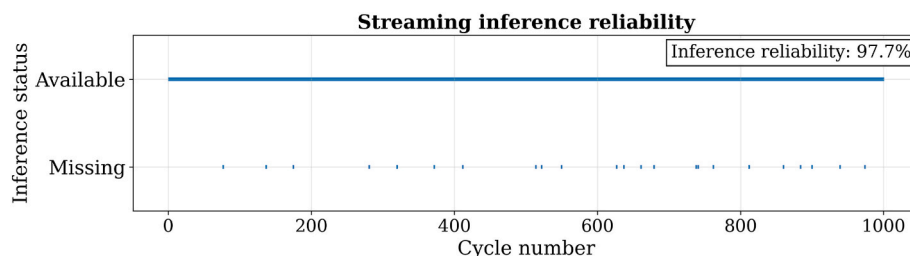


Fig. 12. Streaming inference reliability across 1000 consecutive acquisition cycles.

summarizes the comparative inference-time and memory-footprint results for deep learning, traditional machine learning, and statistical models.

The results indicate that classical models (ARIMA, Random Forest, and XGBoost) exhibit lower inference latency, reflecting their simpler computational structure. However, despite their speed advantage, these models demonstrated inferior prediction accuracy across target variables for time series prediction, where nonlinear dynamics dominate in indoor environment. Among the deep learning models, GRU-SAM achieves a balance between accuracy and computational cost, suitable for real-time edge deployment requirements. In comparison with more complex recurrent architectures, such as BiLSTM and BiLSTM-SAM, GRU-SAM exhibits lower inference latency and reduced peak memory consumption. The peak memory usage of GRU-SAM is comparable to that of classical models, indicating that the proposed attention-enhanced GRU does not impose excessive memory overhead despite its superior predictive performance. These findings demonstrate that GRU-SAM provides the most consistent accuracy–efficiency trade-off, combining high prediction accuracy with moderate inference latency and memory footprint.

5. Conclusions

This study presents the initial phase of the ICTAir framework, an IoT-enabled, deep learning-based system for real-time monitoring and short-term prediction of indoor environmental parameters with moderate inference and memory requirements suitable for edge deployment. Using cost-effective sensors integrated with a cloud platform, the framework continuously monitored temperature, relative humidity, PM_{2.5}, and PM₁₀, with sensor-level calibration significantly improving measurement accuracy. Statistical analysis of field deployment data revealed distinct temporal regimes were observed, with temperature and relative humidity showing low-to-moderate variability and persistence, while PM_{2.5} and PM₁₀ exhibited high variability, positive skewness, and heavy-tailed distributions. Comparative evaluation showed that the proposed GRU-SAM model consistently outperformed classical statistical, machine-learning, and deep-learning baselines, particularly for particulate matter prediction under nonlinear and transient dynamics. The integration of a simplified attention mechanism enhanced forecasting accuracy while maintaining low computational overhead. Optimization analysis identified a 1-h training window and a 5-min prediction horizon as the optimal configuration. Real-time implementation on a Raspberry Pi 4B verified edge feasibility, with the complete data acquisition-to-prediction pipeline consistently satisfying the 1-min latency constraint and achieving 97.7% streaming inference reliability. The GRU-SAM model achieved sub-second inference latency per prediction window, with a mean latency of 750.99 ms, a p95 latency of 716.81 ms, and a peak RAM usage of 605.34 MB, enabling stable end-to-end operation under real-time constraints. Moreover, the deployed GRU-SAM model comprising approximately 13.4k trainable parameters, requiring about 52.5 KB of raw float 32 wt storage, and incurring approximately 1.5 MFLOPs per inference for a 60×1 input window.

This study represents an initial validation conducted within a single laboratory environment using a limited set of four indoor parameters. As such, the generalization of the proposed approach to larger spaces, multi-room layouts, or alternative sensor placements remains a key limitation that requires further investigation. In addition, the current model does not incorporate external influencing factors such as outdoor air quality conditions or real-time occupancy information, which may affect indoor pollutant dynamics and thermal behavior. Despite these limitations, the ICTAir framework establishes a scalable foundation for proactive indoor environmental control. Future work will extend the framework to multi-building deployments, integrate additional pollutants (e.g., CO₂ and VOCs), and enable direct coupling with autonomous control systems for fully closed-loop indoor environment management. In addition, established thermal comfort indices (e.g., PMV/PPD) and

air quality indices will be derived from the predicted environmental parameters, enabling holistic comfort and air quality assessment without compromising real-time deployment constraints.

Credit author statement

Methodology and conceptualization, Jamal Yousaf and Jurng-Jae Yee; investigation and analysis, Jamal Yousaf and Regidestyoko Wasistha Harseno; resources of material, Jamal Yousaf and Jurng-Jae Yee; validation, Jamal Yousaf and Regidestyoko Wasistha Harseno; preparation of the original draft, Jamal Yousaf and Jurng-Jae Yee; editing and reviewing, Jamal Yousaf, Regidestyoko Wasistha Harseno and Jurng-Jae Yee; acquisition of funding and supervision, Jurng-Jae Yee. All authors have read and agreed to the published version of the manuscript.

Declaration of generative AI and AI-assisted technologies in the writing process

During the preparation of this work the authors used ChatGPT in order to improve the language and readability of the text only. After using this tool/service, the authors reviewed the manuscript multiple times and edited the content as needed and took full responsibility for the content of the publication.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgements

This research was supported by the Basic Science Research Program through the National Research Foundation of Korea (NRF) funded by the Ministry of Land, Infrastructure and Transport (RS-2022-00141900).

Data availability

Data will be made available on request.

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